SIMATS ENGINEERING

ASSESSMENT TOOL 3 - ANSWER SHEET

Student Name	Dhakshitha A	Register No.	1925260126
Course Name	Computer Networks	Course Code	CSA0709
Course Outcome	CO2 (BL3): Configure IP addressing schemes, subnetting and basic routing techniques using a simulation tool	Weightage / Tools	30% | Cisco Packet Tracer, Wireshark

Code of Conduct:

I Dhakshitha A / 1925260126 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Criteria	Weightage	How it is addressed in every answer
Problem Identification	20%	Root cause of the fault is stated first
Debugging	25%	Step-by-step calculation / verification
Optimization	20%	Corrected design and configuration
Analysis	20%	Justification and final result
Code Quality	15%	Commented Cisco IOS commands, clean tables

Question 1 - IPv4 Workstation Debugging (10 Marks)

Given: IP 192.168.10.65, mask 255.255.255.192 (/26), gateway 192.168.10.1; the other systems are in 192.168.10.0/26.

1. Problem Identification

A /26 mask gives block size 256 - 192 = **64**, so the subnets of 192.168.10.0 are .0, .64, .128 and .192. The host address .65 lies in the block 64-127, so it belongs to **192.168.10.64/26**, not to 192.168.10.0/26.

Item	192.168.10.0/26 (intended subnet)	192.168.10.64/26 (subnet of the workstation)
Network address	192.168.10.0	192.168.10.64
Broadcast address	192.168.10.63	192.168.10.127
Valid host range	192.168.10.1 - 192.168.10.62	192.168.10.65 - 192.168.10.126
Usable hosts (2^6 - 2)	62	62

2. Debugging - verification

- IP 192.168.10.65: a valid host address *inside 192.168.10.64/26* (first usable host), but it is **outside** the intended subnet 192.168.10.0/26.
- Gateway 192.168.10.1: a valid host address of 192.168.10.0/26, but it is **outside** the workstation's own subnet (192.168.10.64/26). A host can only use a gateway that is on its own subnet, so the gateway is unreachable.
- **Configuration errors:** (a) the IP address is in the wrong subnet, so the workstation cannot reach the other hosts of .0/26 directly; (b) the IP and the gateway are in different subnets (gateway mismatch); the mask itself is correct.

3. Optimization - corrected configuration

Parameter	Corrected value
IP address	192.168.10.10 (any free host in .1 - .62; .1 is reserved for the gateway)
Subnet mask	255.255.255.192 (/26)
Default gateway	192.168.10.1 (router interface, inside 192.168.10.0/26)

```
! Router interface (gateway)
interface GigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.192
 no shutdown
! PC (Packet Tracer > Desktop > IP Configuration)
 IP 192.168.10.10   Mask 255.255.255.192   Gateway 192.168.10.1
! Verify:  ping 192.168.10.1   |   ping 192.168.10.20
```

4. Analysis / Result

After correction, the workstation, the gateway and all peers are in 192.168.10.0/26. **Usable host addresses = 2^6 - 2 = 64 - 2 = 62** (1 used by the gateway, 61 left for end devices).

Question 2 - IPv6 Configuration Debugging (10 Marks)

Given: Device A = 2001:db8::ff00:42:8329/64 and Device B = 2001:db8:0:0:1::1/64; same network but communication fails.

1. Expansion of the addresses (8 groups of 16 bits)

Device	Compressed	Full notation
A	2001:db8::ff00:42:8329	2001:0db8:0000:0000:0000:ff00:0042:8329 (:: = 3 zero groups)
B	2001:db8:0:0:1::1	2001:0db8:0000:0000:0001:0000:0000:0001 (:: = 2 zero groups)

2. Network prefix (first 64 bits = first 4 groups)

Device	Network prefix (/64)	Interface ID (last 64 bits)
A	2001:0db8:0000:0000 = 2001:db8::/64	0000:ff00:0042:8329
B	2001:0db8:0000:0000 = 2001:db8::/64	0001:0000:0000:0001

3. Same-subnet check and problem identification

Both devices have the identical prefix **2001:db8::/64**, so by calculation they **are in the same subnet**; there is no real prefix-length mismatch (both are /64). The fault is therefore in the way the addresses were written/assigned:

- Device B contains a non-zero group (**0001**) in the 5th position. This looks like an attempt to use a subnet ID such as 2001:db8:0:1::/64, but with /64 it falls inside the interface ID. A mistyped extra `:0` is the likely cause; the two addresses follow no common design.
- If B was meant to be in subnet 2001:db8:0:1::/64, A and B would be in *different* subnets and need a router - that is the mismatch the administrator intended to avoid.
- With the prefix correct, remaining causes to check are the wrong VLAN/port, a blocked ICMPv6 (Neighbor Discovery), IPv6 routing not enabled, or no gateway.

4. Optimization - corrected plan

Device	Corrected address	Remarks
Router (gateway)	2001:db8:0:1::1/64	Subnet 2001:db8:0:1::/64
Device A	2001:db8:0:1::10/64	Clean, easy-to-read host ID
Device B	2001:db8:0:1::20/64	Same 4-group prefix as A

```
interface GigabitEthernet0/0
 ipv6 address 2001:DB8:0:1::1/64
 no shutdown
ipv6 unicast-routing
! Device A: 2001:db8:0:1::10/64  gw 2001:db8:0:1::1
! Device B: 2001:db8:0:1::20/64  gw 2001:db8:0:1::1
```

5. Analysis / Result

A /64 leaves 128 - 64 = 64 host bits. **Available host addresses = 2^{64} = 18,446,744,073,709,551,616** (IPv6 has no broadcast address; excluding the subnet-router anycast address it is 2^{64} - 1).

Question 3 - Subnet Allocation Analysis and VLSM Optimization (10 Marks)

Given: 192.168.1.0/24 split into /26, /27 and /28 subnets. The question gives no department sizes, so the following **assumed requirements** are used to demonstrate the method: Engineering 20 hosts, Sales 45, HR 25, Finance 10.

1. Existing allocation - addresses, usable and unused

Dept	Mask	Network	Broadcast	Host range	Usable	Need	Unused / short
Engineering	/26	192.168.1.0	192.168.1.63	.1 - .62	62	20	42 unused
Sales	/27	192.168.1.64	192.168.1.95	.65 - .94	30	45	15 SHORT
HR	/28	192.168.1.96	192.168.1.111	.97 - .110	14	25	11 SHORT
Finance	/28	192.168.1.112	192.168.1.127	.113 - .126	14	10	4 unused
Free block	/25	192.168.1.128	192.168.1.255	.129 - .254	126	-	128 addresses idle

Usable hosts = 2^n - 2 (n = host bits: /26 = 6, /27 = 5, /28 = 4).

2. Problem identification and debugging

- **Inefficient allocation:** the largest block (/26) went to Engineering which needs only 20 hosts - 42 addresses wasted.
- **Mistake - undersized subnets:** Sales (needs 45, has 30) and HR (needs 25, has 14) are short, which causes the shortage complaints.
- **Mistake - no VLSM order:** blocks were chosen without matching size to demand (largest need should get the largest block), and a whole /25 (128 addresses) lies unused.
- All four subnets start on correct boundaries and do not overlap; the faults are sizing faults, not overlap faults.

3. Optimization - VLSM scheme (largest requirement first)

Dept	Need	Mask	Network	Host range	Broadcast	Usable	Spare
Sales	45	/26 (255.255.255.192)	192.168.1.0	.1 - .62	192.168.1.63	62	17
HR	25	/27 (255.255.255.224)	192.168.1.64	.65 - .94	192.168.1.95	30	5
Engineering	20	/27 (255.255.255.224)	192.168.1.96	.97 - .126	192.168.1.127	30	10
Finance	10	/28 (255.255.255.240)	192.168.1.128	.129 - .142	192.168.1.143	14	4
WAN link	2	/30 (255.255.255.252)	192.168.1.144	.145 - .146	192.168.1.147	2	0

Analysis: every department now fits its requirement. Only 64 + 32 + 32 + 16 + 4 = 148 addresses are consumed (192.168.1.0 - .147) and **192.168.1.148 - 192.168.1.255 (108 addresses)** stays free for growth, compared with unmet demand in the old plan.

```
! Example: Sales gateway and Finance gateway on the router
interface GigabitEthernet0/0.10
 ip address 192.168.1.1 255.255.255.192     ! Sales
interface GigabitEthernet0/0.40
 ip address 192.168.1.129 255.255.255.240    ! Finance
```

Question 4 - IPv6 Configuration Debugging (10 Marks)

This question has the same addresses as Question 2; the complete working is given again with simulator verification.

Given: Device A = 2001:db8::ff00:42:8329/64 and Device B = 2001:db8:0:0:1::1/64; same network but communication fails.

1. Expansion of the addresses (8 groups of 16 bits)

Device	Compressed	Full notation
A	2001:db8::ff00:42:8329	2001:0db8:0000:0000:0000:ff00:0042:8329 (:: = 3 zero groups)
B	2001:db8:0:0:1::1	2001:0db8:0000:0000:0001:0000:0000:0001 (:: = 2 zero groups)

2. Network prefix (first 64 bits = first 4 groups)

Device	Network prefix (/64)	Interface ID (last 64 bits)
A	2001:0db8:0000:0000 = 2001:db8::/64	0000:ff00:0042:8329
B	2001:0db8:0000:0000 = 2001:db8::/64	0001:0000:0000:0001

3. Same-subnet check and problem identification

Both devices have the identical prefix **2001:db8::/64**, so by calculation they **are in the same subnet**; there is no real prefix-length mismatch (both are /64). The fault is therefore in the way the addresses were written/assigned:

- Device B contains a non-zero group (**0001**) in the 5th position. This looks like an attempt to use a subnet ID such as 2001:db8:0:1::/64, but with /64 it falls inside the interface ID. A mistyped extra `:0` is the likely cause; the two addresses follow no common design.
- If B was meant to be in subnet 2001:db8:0:1::/64, A and B would be in *different* subnets and need a router - that is the mismatch the administrator intended to avoid.
- With the prefix correct, remaining causes to check are the wrong VLAN/port, a blocked ICMPv6 (Neighbor Discovery), IPv6 routing not enabled, or no gateway.

4. Optimization - corrected plan

Device	Corrected address	Remarks
Router (gateway)	2001:db8:0:1::1/64	Subnet 2001:db8:0:1::/64
Device A	2001:db8:0:1::10/64	Clean, easy-to-read host ID
Device B	2001:db8:0:1::20/64	Same 4-group prefix as A

```
interface GigabitEthernet0/0
 ipv6 address 2001:DB8:0:1::1/64
 no shutdown
ipv6 unicast-routing
! Device A: 2001:db8:0:1::10/64  gw 2001:db8:0:1::1
! Device B: 2001:db8:0:1::20/64  gw 2001:db8:0:1::1

! Verification in Packet Tracer / Wireshark
show ipv6 interface brief
ping 2001:db8:0:1::20          (from Device A)
! Wireshark filter: icmpv6  -> Neighbor Solicitation / Advertisement
!                   must be seen between A and B
```

5. Analysis / Result

A /64 leaves 128 - 64 = 64 host bits. **Available host addresses = 2^{64} = 18,446,744,073,709,551,616** (IPv6 has no broadcast address; excluding the subnet-router anycast address it is 2^{64} - 1).

Question 5 - Routing Table Debugging (10 Marks)

Given routing table: 192.168.1.0/24 via 10.0.0.2; 192.168.2.0/24 via 10.0.0.3; default route via 10.0.0.1. Packets for 192.168.2.100 do not reach the intended network.

1. Longest-prefix match

Route	Range covered	Matches 192.168.2.100?	Prefix length
192.168.1.0/24	192.168.1.0 - 192.168.1.255	No	-
192.168.2.0/24	192.168.2.0 - 192.168.2.255	**Yes**	**/24**
Default (0.0.0.0/0)	All addresses	Yes	/0

Two routes match; the router chooses the **longest prefix (/24 beats /0)**. The selected entry is 192.168.2.0/24 and the next-hop router is **10.0.0.3**.

2. Subnet verification and debugging

192.168.2.100 lies in 192.168.2.0/24 (network .2.0, broadcast .2.255, hosts .2.1 - .2.254), so it does belong to the intended subnet, and the table-lookup logic is correct. The fault is therefore not in prefix matching but in the table data, which has these weaknesses:

- The next hops of 192.168.1.0/24 (10.0.0.2) and 192.168.2.0/24 (10.0.0.3) may be **interchanged**; if 10.0.0.3 is not the router attached to 192.168.2.0/24, packets go to the wrong router and are dropped.
- No **exit interface** and no directly connected route (10.0.0.0/x) is listed, so reachability of 10.0.0.3 is not guaranteed.
- The default route is shown only as a name (not 0.0.0.0/0), and no administrative distance or backup path is defined.

Check: run `show ip route` and `ping 10.0.0.3`; use Packet Tracer simulation mode to see where the packet is dropped.

3. Optimization - corrected routing table

Destination	Next hop / exit	Remarks
192.168.2.0/24	10.0.0.3 via Gi0/0	Verified next hop for the network that holds 192.168.2.100
192.168.1.0/24	10.0.0.2 via Gi0/0	Specific route kept
0.0.0.0/0	10.0.0.1 via Gi0/0	Gateway of last resort

```
ip route 192.168.1.0 255.255.255.0 10.0.0.2
ip route 192.168.2.0 255.255.255.0 10.0.0.3
ip route 0.0.0.0 0.0.0.0 10.0.0.1
! Optional backup (floating) default route, higher distance
ip route 0.0.0.0 0.0.0.0 10.0.0.4 200
! Verify:  show ip route   |   traceroute 192.168.2.100
```

The two /24 routes use different next hops, so they cannot be merged (summarised); summarisation is applicable only when several networks share the same next hop. Specific routes with exit interfaces avoid recursive lookups and speed up forwarding.

4. Analysis / Result

- Packet for 192.168.2.100 -> matches 192.168.2.0/24 (longest prefix) -> forwarded to **10.0.0.3**.
- If no destination entry matches (for example 8.8.8.8 or 192.168.3.5), the **default route 0.0.0.0/0 via 10.0.0.1** is used.